# Parametric & Nonparametric ML models

## Parametric ML models

- Parametric and non-parametric models are two fundamental approaches in machine learning, each with distinct characteristics, advantages, and limitations.
- Understanding their differences is crucial for selecting the right model for a given problem.
- Parametric models assume a **specific functional form for the relationship between input and output variables**.
- They summarize data using a fixed number of parameters, regardless of the dataset size.
- **Examples include** linear regression, logistic regression, and Naive Bayes.

## Parametric ML models (cntd..)

- These models are **computationally efficient and require less data to train.** However, they are constrained by their assumptions, making them less flexible and potentially unsuitable for capturing complex relationships in data.
- For instance, linear regression assumes a linear relationship, which may not align with the true underlying function.

**Example**

Linear regression models the relationship as:

```
y = b0 + b1*x1 + b2*x2
```

Here, b0, b1, and b2 are fixed parameters learned from the data.

## Non Parametric ML models

- Non-parametric models **do not assume a predefined functional form**.
- They are highly flexible and can **adapt to complex patterns in data**.
- **Examples include** k-Nearest Neighbors (KNN), decision trees, and support vector machines.

- These models require more data and computational resources.
- They are prone to overfitting but can achieve higher accuracy when sufficient data is available.
- Unlike parametric models, they are less interpretable.

## Non Parametric ML models

### Example

KNN predicts the output for a new data point based on the majority class of its k nearest neighbors.

```
from sklearn.neighbors import KNeighborsClassifier



model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

## Parametric vs Non Parametric ML models

- **Assumptions**: Parametric models make **strong assumptions about data** distribution, while non-parametric models make minimal or no assumptions.
- **Flexibility**: Parametric models are less flexible, whereas non-parametric models can adapt to complex data structures.
- **Data Requirements**: Parametric models require less data, while non-parametric models need larger datasets to perform well.
- **Interpretability**: Parametric models are more interpretable, making them suitable for inference tasks. Non-parametric models are better for prediction accuracy.

## Usage of Parametric vs Non Parametric ML models

•Use **parametric models** when the data aligns with the model's assumptions, or when interpretability and computational efficiency are priorities.

•Use **non-parametric models** for complex, high-dimensional data where flexibility and accuracy are more critical than interpretability.

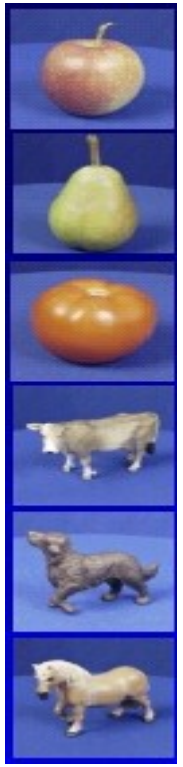
- Understanding these distinctions ensures the selection of an appropriate model, balancing accuracy, interpretability, and computational efficiency.

# Overfitting and Underfitting

# Generalization



Training set (labels known)



Test set (labels unknown)

- How well does a learned model generalize from the data it was trained on to a new test set?

Slide credit: L. Lazebnik

# No Free Lunch Theorem




Slide credit: D. Hoiem

## What is overfitting?

- In machine learning, overfitting is a phenomenon that occurs when a machine learning algorithm begins to memorize the training data rather than learning the underlying patterns.
- It occurs when a model is too complex for the data it is supposed to be modeling.
- This can lead to poor performance on new data, **as the algorithm is not able to generalize from the training data to other datasets.**
- Overfitting is a common problem in machine learning and can be caused by a variety of factors.

## What is underfitting?

•Underfitting, on the other hand, occurs when a model is not complex enough. While there can be different causes, it is often the case that the **model is not given enough data to learn from.**

•As a result, the model is **not able to learn the generalizable patterns in the data** and ends up performing poorly on both the training dataset and new data points.

## What is a good fit in machine learning?

A good fit in machine learning is defined as a model that accurately predicts the output of new data.

The goal of machine learning is to build models that generalize well, which means they have a low error rate on a test set with unseen data.

# Bias and Variance

•Error of a model can be split into reducible and irreducible.

Reducible:

•bias (“error due to squared bias”)

•variance (“error due to variance”)

Irreducible: the error we cannot remove with our model caused by things outside of our control, like statistical noise in observations (also called irreducible noise).

•Bias — is a difference between model predictions and training data on one set.

•Variance — difference between model predictions and expected value over all training sets.

# Understanding Bias

- Mathematically , Bias=Actual Value-Predicted Value

**Example**
Suppose we have the following training data:

| Temperature (°F) | Actual Yield |
|---|---|
| 50 | 3.3 |
| 70 | 2.3 |
| 90 | 3.0 |

Assume the model predicts:

| Temperature | Actual | Predicted |
|---|---|---|
| 50 | 3.3 | 2.9 |
| 70 | 2.3 | 2.7 |
| 90 | 3.0 | 2.8 |

# Bias

**High Bias**

A model with high bias:

- Is too simple.
- Cannot learn the true relationship.
- Makes systematic errors.
- **Usually underfits the data**.

Example:

- Fitting a straight line to data that actually follow a quadratic curve.

# Understanding Variance

- Variance measures how much the learned model changes when the training data change.

**High Variance**

```
Dataset 1  -----------
Dataset 2         /
Dataset 3       \
```

- Every dataset gives a different regression line.
- Model is unstable.

# Bias vs Variance

| Aspect | Bias | Variance |
| --- | --- | --- |
| Definition | Difference between the model's predictions and the actual values because the model is too simple. | Difference in the model's predictions when trained on different training datasets. |
| Question it Answers | How accurate are the model's predictions? | How stable is the model when the training data change? |
| Cause | Model is too simple and cannot capture the true relationship. | Model is too sensitive to small changes in the training data. |
| Training Error | High | Low |

# Bias vs Variance key difference

## Key Difference Between Bias and Variance

| Feature | Bias | Variance |
|---|---|---|
| Definition | Difference between the model's predicted values and the actual values due to an overly simple model. | Difference in the model's predictions when trained on different training datasets. |
| Measures | Accuracy of the model. | Stability (consistency) of the model. |
| Cause | Model is too simple. | Model is too complex. |
| Effect on Model | Underfits the data. | Overfits the data. |
| Training Error | High | Low |
| Testing Error | High | High |
| Sensitivity to Training Data | Low | High |
| Learning Ability | Fails to capture the true relationship in the data. | Captures both the true pattern and the noise in the data. |
| Model Complexity | Low | High |
| Prediction Performance | Poor on both training and testing data. | Excellent on training data but poor on unseen testing data. |
| Example (Simple Linear Regression) | Fitting a straight line to data that actually follow a quadratic relationship. | Regression line changes significantly when the training dataset changes slightly. |
| How to Reduce | Increase model complexity or add more relevant features. | Simplify the model, collect more training data, or use regularization. |
| Associated Problem | Underfitting | Overfitting |

# Polynomial regression

# Polynomial regression

What is Polynomial Regression?

- Polynomial Regression is a form of. *regression analysis in which the relationship between the independent variables and dependent variables are modeled in the nth degree polynomial.*

- Polynomial Regression is a special case of Linear Regression where we *fit the polynomial equation on the data with a curvilinear relationship between the dependent and independent variables*.

- A Quadratic Equation is a Polynomial Equation of 2nd Degree. However, this degree can increase to nth values.

Polynomial regression

# Types of Polynomials

Linear —— $ax + b = 0$

Quadratic —— $ax^2 + bx + c = 0$

Cubic —— $ax^3 + bx^2 + cx + d = 0$

# Polynomial regression

## Assumptions of Polynomial Regression:

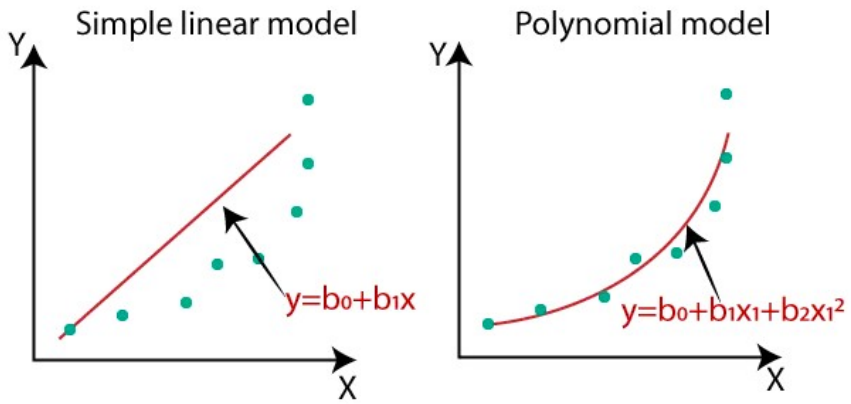
- The behavior of a dependent variable can be explained by a linear, or curvilinear, additive relationship between the dependent variable and a set of *k* independent variables *(xi, i=1 to k)*.

- The relationship between the dependent variable and any independent variable is linear or curvilinear (specifically polynomial).

- The independent variables are independent of each other.

- The errors are independent, normally distributed with mean zero and a constant variance (*OLS*).

# Polynomial regression




**Why do we need Polynomial Regression?**

Let's consider a case of ***Simple Linear Regression***.

•We make our model and find out that it performs very badly,

•We observe the distance between the actual value and the best fit line, which we predicted and it seems that the **actual value has some kind of curve in the graph** and our line is no where near to cutting the mean of the points.

• This is where polynomial Regression comes to the play, it predicts the **best-fit line that follows the pattern(curve) of the data,** as shown in the picture

## Polynomial regression

- Polynomial Regression *does not require the relationship between the independent and dependent variables to be linear in the data set*, This is also one of the main difference between the Linear and Polynomial Regression.

- **Polynomial Regression is generally used when the points in the data are not captured by the Linear Regression Model and the Linear Regression fails in describing the best result clearly**.

As we increase the degree in the model, it tends to increase the performance of the model. However, increasing the degrees of the model also increases the risk of **over-fitting and under-fitting** the data.

# Polynomial regression

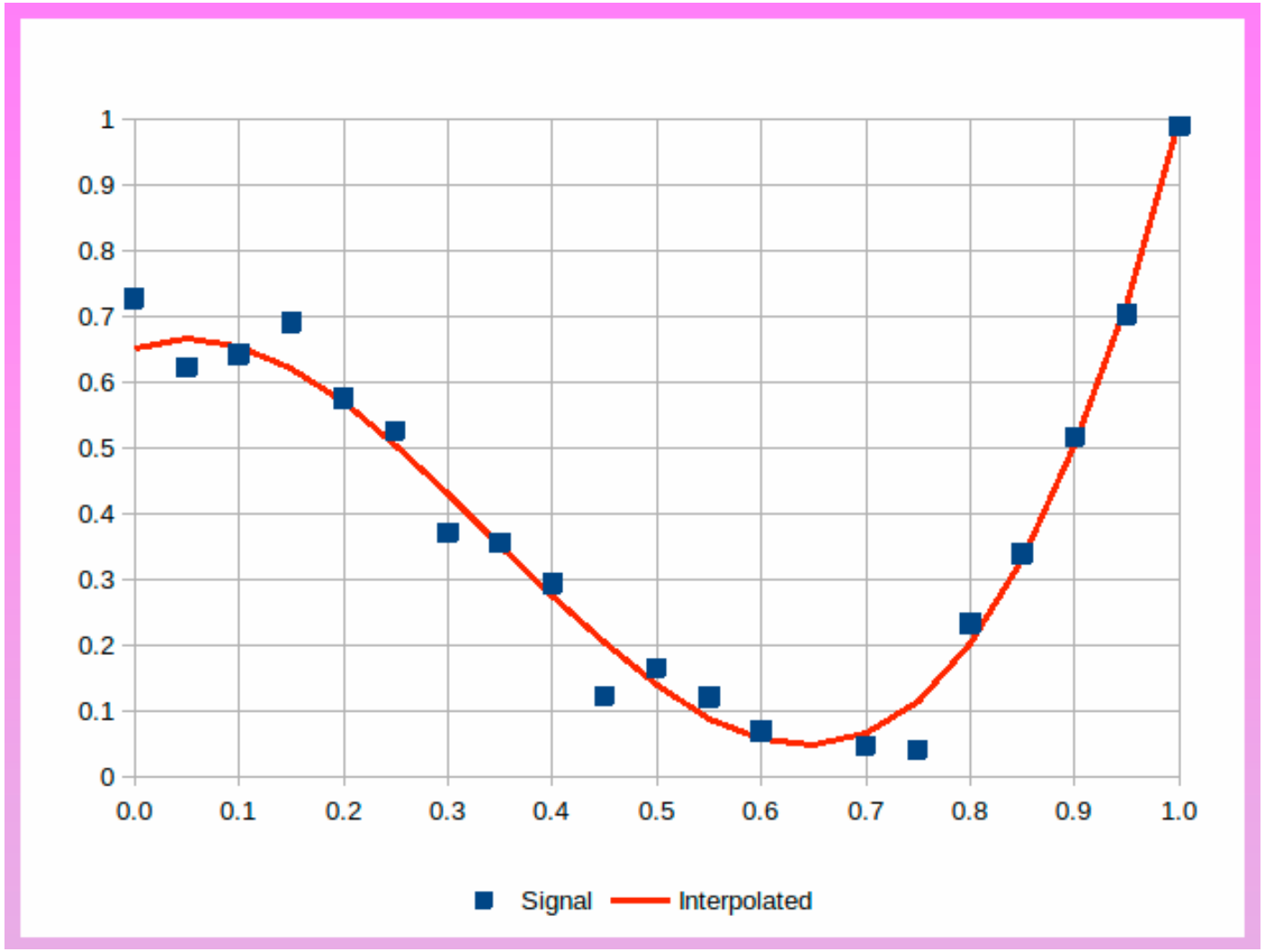
**How to find the right degree of the equation?**
In order to find the right degree for the model to prevent over-fitting or under-fitting, we can use:

**1.Forward  Selection:**
This method increases the degree until it is significant enough to define the best possible model.

**2.Backward Selection:**
This method decreases the degree until it is significant enough to define the best possible model.

# Polynomial regression

Simply put polynomial regression is an attempt to create a polynomial function that approximates a set of data points. This is easier to demonstrate with a visual example.



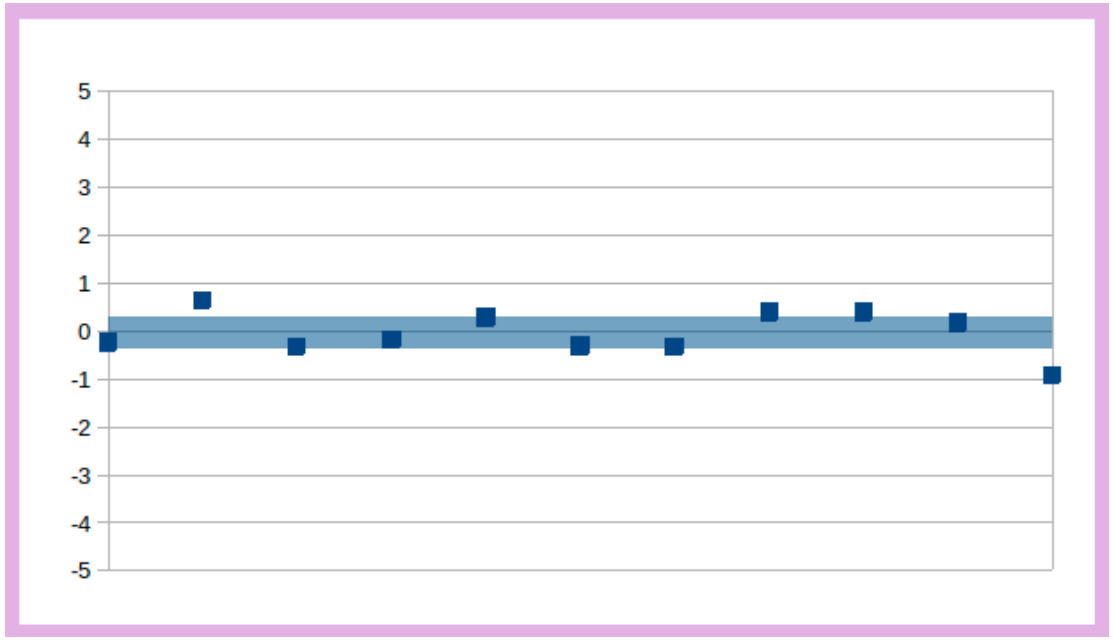
# Polynomial regression

- Polynomial regression is one of several methods of **curve fitting**.
- With polynomial regression, the data is approximated using a polynomial function.
- A polynomial is a function that takes the form $f(x) = c_0 + c_1 x + c_2 x^2 \cdots c_n x^n$

  where *n* is the degree of the polynomial and *c* is a set of coefficients.

A polynomial of degree 0 is just a constant because $f(x) = c_0 x^0 = c_0$.

Likewise preforming polynomial regression with a degree of 0 on a set of data returns a single constant value. It is the same as the mean average of that data. This makes sense because the average is an approximation of all the data points.

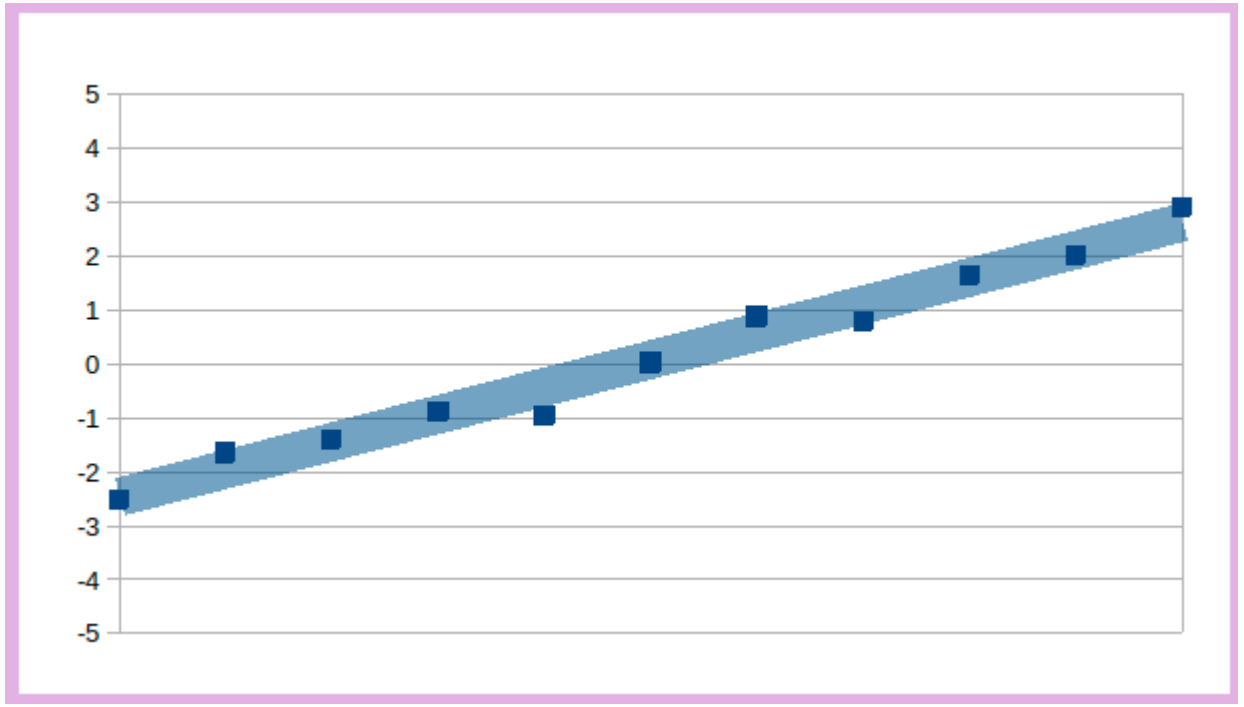


# Polynomial regression

**Linear regression** is polynomial regression of degree 1, and generally takes the form

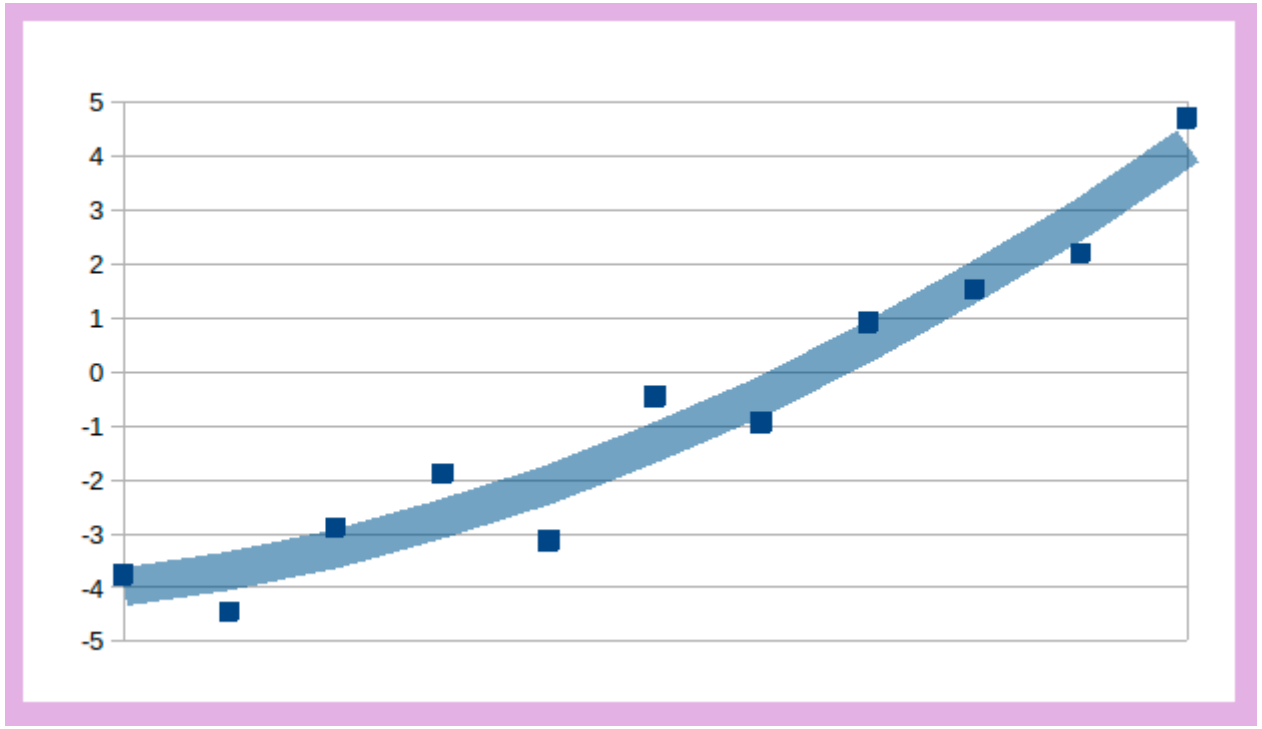
$y = m\,x + b$ where $m$ is the slope, and $b$ is the y-intercept.

It could just as easily by written $f(x) = c_0 + c_1\,x$ with $c_1$ being the slope and $c_0$ the y-intercept.



# Polynomial regression

Quadratic regression is a $2^{nd}$ degree polynomial and not nearly as common. Now the regression becomes non-linear and the data is not restricted to straight lines.



## Polynomial regression

We need a way of calculating coefficients that takes all our data points into consideration. This is where least squares come in.

To preform a least square approach we need to define a **residual** function.

For any point $(x_i, y_i)$ on the line, the residual would be:

**Error or** $$r_i(x_i) = y_i - (mx_i + b)$$

Where $i$ is the index into the set of known data points.

We can generalize it for any function:

**Error or** $$r_i(x_i) = y_i - f(x_i)$$

# Polynomial regression

*What does the residual tell us?*

Well, for each data point the *residual denotes the amount of error between our estimation and the true data*.

*How about the overall error*?

To get this, we could just *sum up the error for all data points*.

However error can be *positive or negative*.

So we could end up with a lot of error uniformly distributed between negative and positive values.

$$r(x) = \sum_{i=0}^{n} \left[ y_i - f(x_i) \right]^2$$

So now that we have a function that measures residual, what do we do with it?

Well, if we are trying to produce a function that models a set of data, we want the residual to be as small as possible—we want to **minimize** it.

## Polynomial regression

Let's try using a 2nd degree polynomial to get the results for quadratic regression.

First, the quadratic function:

$$y = c_0 + c_1 x + c_2 x^2$$

Now the sum of squares for n known data points:

$$r(x) = \sum_{i=0}^{n} (c_0 + c_1 x_i + c_2 x^2 - y_i)^2$$

# Polynomial regression

**Examples** Find a quadratic regression model for the following data:

| $x$ | $y$ |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 15 |

# Polynomial regression

The polynomial equation in given by $\boldsymbol{y = a_0 + a_1 x + a_2 x^2}$

The coefficients $\boldsymbol{a_0, a_1}$ ***and*** $\boldsymbol{a_2}$ are calculated using the formula,

$$\boldsymbol{a = X^{-1} B}$$

The values of a0, a1, and a2 are calculated using the following system of equations:

$$\sum y_i = n a_0 + a_1 \left(\sum x_i\right) + a_2 \left(\sum x_i^2\right)$$

$$\sum y_i x_i = a_0 \left(\sum x_i\right) + a_1 \left(\sum x_i^2\right) + a_2 \left(\sum x_i^3\right)$$

$$\sum y_i x_i^2 = a_0 \left(\sum x_i^2\right) + a_1 \left(\sum x_i^3\right) + a_2 \left(\sum x_i^4\right)$$

# Polynomial regression

**Examples** Find a quadratic regression model for the following data:

| x | y |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 15 |

Consider the polynomial of 2nd degree.

The polynomial equation in given by $y = a_0 + a_1 x + a_2 x^2$

The coefficients $a_0, a_1$ *and* $a_2$ are calculated using the formula,

$$a = X^{-1} Y$$

Where,

$$X = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad Y = \begin{bmatrix} \sum y_i \\ \sum (x_i, y_i) \\ \sum (x_i^2, y_i) \end{bmatrix}$$

# Polynomial regression

| x | y |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 15 |

| $x_i$ | $y_i$ | $x_i y_i$ | $x_i^2$ | $x_i^2 y$ | $x_i^3$ | $x_i^4$ |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 4 | 8 | 4 | 16 | 8 | 16 |
| 3 | 9 | 27 | 9 | 81 | 27 | 81 |
| 4 | 15 | 60 | 16 | 240 | 64 | 256 |
| $\Sigma x_i = 10$ | $\Sigma y_i = 29$ | $\Sigma x_i y_i = 96$ | $\Sigma x_i^2 = 30$ | $\Sigma x_i^2 y_i = 338$ | $\Sigma x_i^3 = 100$ | $\Sigma x_i^4 = 354$ |

$$X = \begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \quad Y = \begin{bmatrix} \Sigma y_i \\ \Sigma (x_i, y_i) \\ \Sigma (x_i^2, y_i) \end{bmatrix}$$

$$a = X^{-1} Y$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix}$$

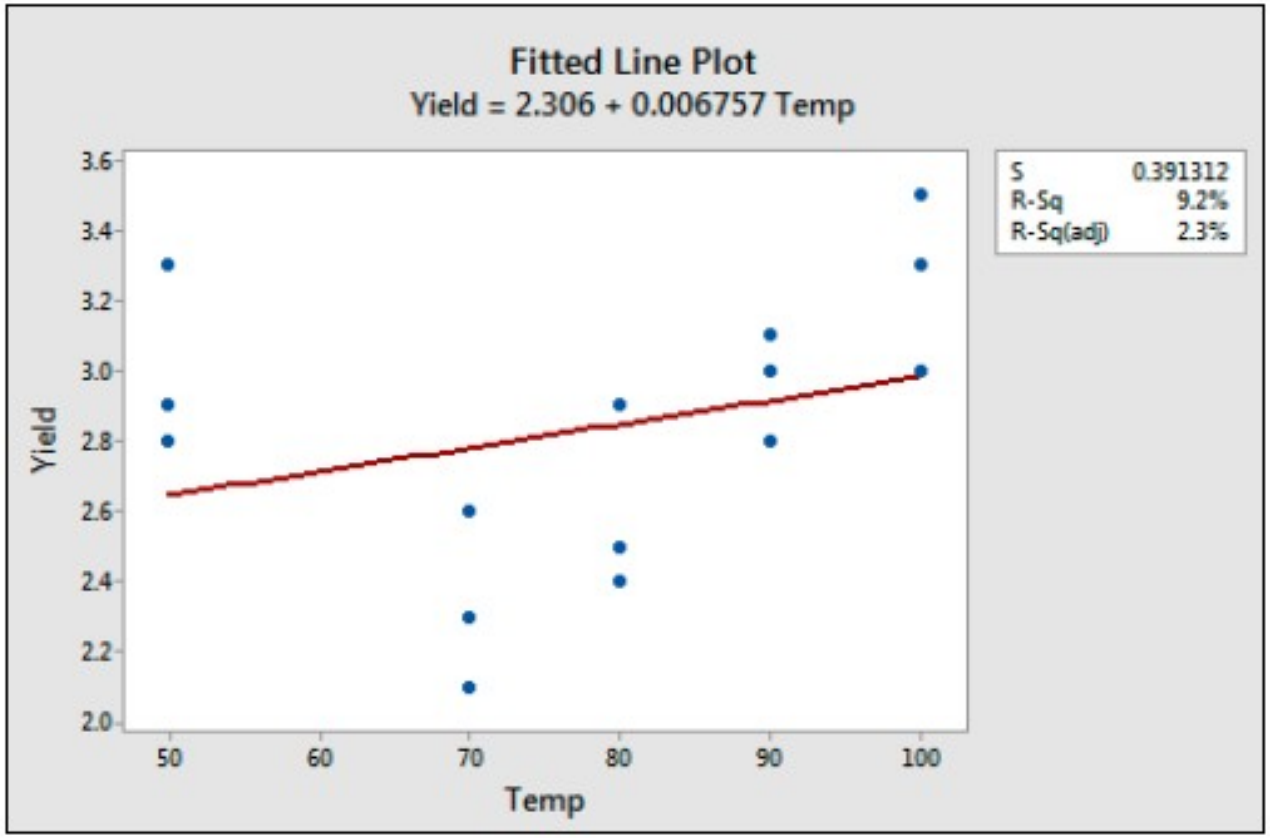
$$y = -0.75 + 0.95\,x + 0.75\,x^2$$

# Polynomial regression

*Example : Yield Data Set-This data set of size n = 15 contains measurements of yield from an experiment done at five different temperature levels. The variables are y = yield and x = temperature in degrees Fahrenheit. The table below gives the data used for this analysis. Find a regression model.*

| i | Temperature | Yield |
|---|---|---|
| 1 | 50 | 3.3 |
| 2 | 50 | 2.8 |
| 3 | 50 | 2.9 |
| 4 | 70 | 2.3 |
| 5 | 70 | 2.6 |
| 6 | 70 | 2.1 |
| 7 | 80 | 2.5 |
| 8 | 80 | 2.9 |
| 9 | 80 | 2.4 |
| 10 | 90 | 3.0 |
| 11 | 90 | 3.1 |
| 12 | 90 | 2.8 |
| 13 | 100 | 3.3 |
| 14 | 100 | 3.5 |
| 15 | 100 | 3.0 |

The figures below give a scatterplot of the raw data and then another scatterplot with lines pertaining to a linear fit and a quadratic fit overlayed. Obviously the trend of this data is better suited to a quadratic fit.

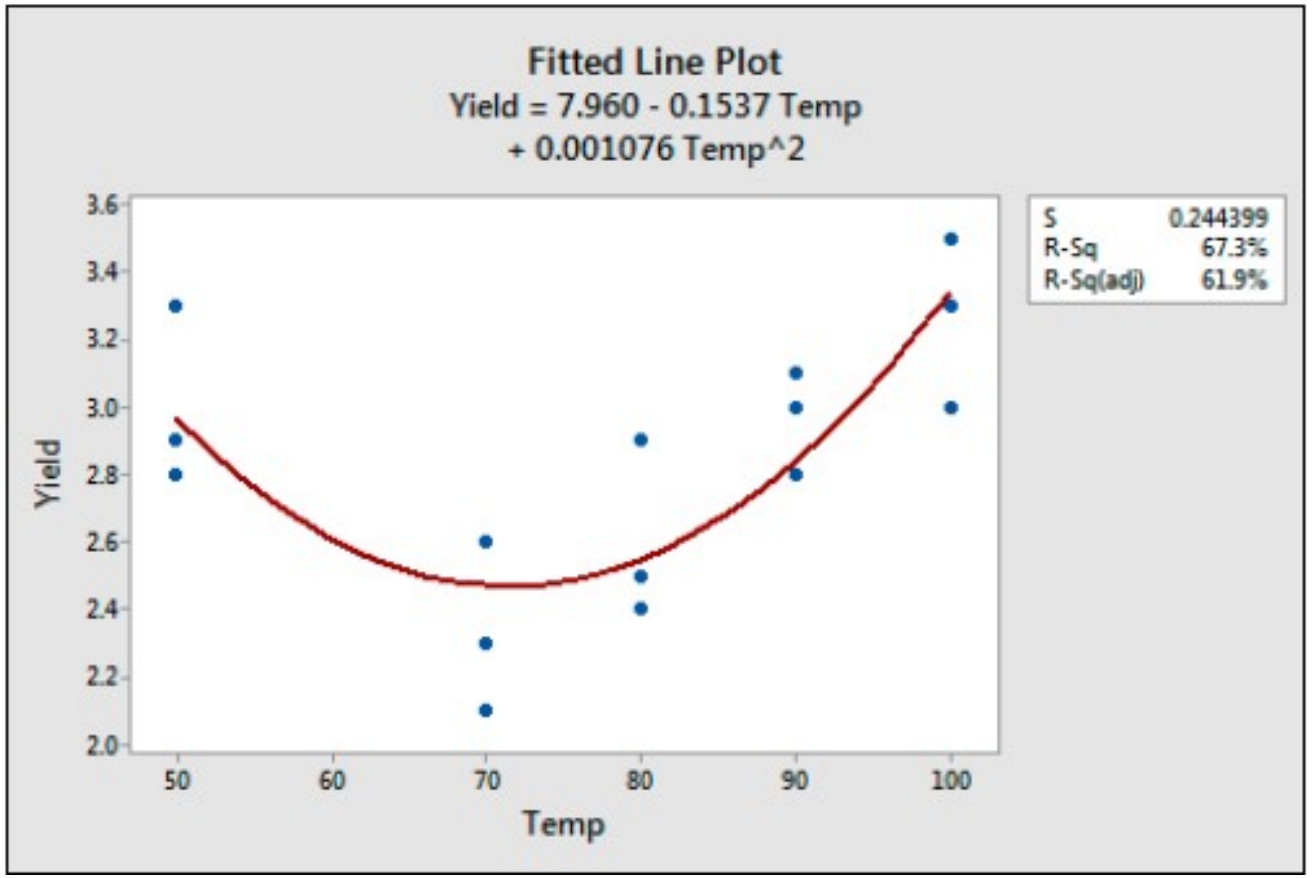


# Polynomial regression

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| 8 | 80 | 2.9 |
| 9 | 80 | 2.4 |
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| 15 | 100 | 3.0 |

The figures below give a scatterplot of the raw data and then another scatterplot with lines pertaining to a linear fit and a quadratic fit overlayed. Obviously the trend of this data is better suited to a quadratic fit.



## Polynomial regression

Let there be only one independent variable x and the relationship between x, and dependent variable y, be modeled as,

$y=a0+a1*x+a2*x2 +.....+ +an*xn$

for some positive integer n >1, then we have a polynomial regression.

Problem Definition:

Find a quadratic regression model for the following data:

| X | Y |
|---|---|
| 3 | 2.5 |
| 4 | 3.2 |
| 5 | 3.8 |
| 6 | 6.5 |
| 7 | 11.5 |

## Polynomial regression

Solution:

Let the quadratic polynomial regression model be

y=a0+a1*x+a2*x2

The values of a0, a1, and a2 are calculated using the following system of equations:

$$\sum y_i = na_0 + a_1\left(\sum x_i\right) + a_2\left(\sum x_i^2\right)$$

$$\sum y_i x_i = a_0\left(\sum x_i\right) + a_1\left(\sum x_i^2\right) + a_2\left(\sum x_i^3\right)$$

$$\sum y_i x_i^2 = a_0\left(\sum x_i^2\right) + a_1\left(\sum x_i^3\right) + a_2\left(\sum x_i^4\right)$$

## Polynomial regression

First, we calculate the required variables and note them in the following table.

| | x | y | x2 | x3 | x4 | y*x | y*x2 |
|---|---|---|---|---|---|---|---|
| | 3 | 2.5 | 9 | 27 | 81 | 7.5 | 22.5 |
| | 4 | 3.2 | 16 | 64 | 256 | 12.8 | 51.2 |
| | 5 | 3.8 | 25 | 125 | 625 | 19 | 95 |
| | 6 | 6.5 | 36 | 216 | 1296 | 39 | 234 |
| | 7 | 12 | 49 | 343 | 2401 | 80.5 | 563.5 |
| Σ | 25 | 27.5 | 135 | 775 | 4659 | 158.8 | 966.2 |

Using the given data we,

$$27.5 = 5a_0 + 25a_1 + 135a_2$$

$$158.8 = 25a_0 + 135a_1 + 775a_2$$

$$966.2 = 135a_0 + 775a_1 + 4659a_2$$

# Polynomial regression

Solving this system of equations we get

$a_0$=12.4285714

$a_1$=-5.5128571

$a_2$=0.7642857

The required quadratic polynomial model is

$$\mathbf{y=12.4285714 -5.5128571 * x +0.7642857 * x^2}$$